Ananya Deepak Deoghare

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Summary

Senior Associate Algorithm Engineer specializing in the architecture and scale of revenue-driving ranking systems, combining published research expertise with production-grade ML infrastructure to deliver multi-million dollar business impact.

Technical Skills

- **Tools & Languages:** Python, SQL, Git, Cython, AWS, GCP, MATLAB, Java, C, C++, HTML, Visual Studio, JIRA, Perforce
- **Databases:** MySQL, Amazon Redshift, Apache Hive, Feature Stores, Data Pipelines
- **Libraries:** Scikit-learn, PyTorch, TensorFlow, NumPy, Pandas, SHAP, OpenCV, Scipy, Cython
- **ML Expertise:** Ranking & Recommendations, CTR/Conversion Modeling, A/B Testing, Feature Engineering, Multi Armed Bandits, Model Monitoring, MLOps

Experience

Senior Associate Algorithm Engineer Jun 2023 – Present
Quinstreet *Foster City, CA*

- Designed and optimized machine learning models for predictive modeling and ranking on a large-scale personal loans user dataset; performed in-depth exploratory data analysis to uncover behavioral patterns and drive feature strategy.
- Built and productionized Gradient Boosting Machine (GBM) models with rigorous hyperparameter tuning, cross-validation, and feature selection, improving ranking quality and predictive performance in the banking domain.
- Applied clustering and controlled A/B experimentation to optimize advertiser click-through rates, driving a projected 10% increase in user engagement.
- Led weekly model diagnostics and performance analysis on user interaction data, increasing model accuracy by 5% and reducing error rates by 10%.
- Engineered high-signal features through systematic feature extraction and selection, significantly improving model efficiency, interpretability, and generalization.
- Partnered with data engineering and product teams to deploy SEO-driven ML models to production, enabling real-time decision-making in high-impact business workflows.
- Implemented continuous model evaluation pipelines with automated hyperparameter optimization and validation frameworks to ensure robustness and alignment with business KPIs.
- Own the performance and reliability of the team's Cython simulation framework used to stress-test ranking strategies before release; optimized processing time by 50%, meeting production-grade performance requirements.
- Implemented the Hungarian Algorithm for optimal ranking assignment, benchmarking against a greedy baseline and increasing margin revenue by 5–8%; improved handling of edge cases missed by heuristic approaches.
- Technical owner of the end to end ML ranking architecture for the Personal Loans vertical (multi-million-dollar revenue line); scope expanded in 2026 to include the conversion model alongside CTR models, owning ranking performance, deployment strategy, experimentation, and long-term system scalability.
- Built and productionized a conversion prediction model for the Personal Loans funnel, complementing CTR models to optimize downstream conversion.

Machine Learning Intern Jun 2022 – Sep 2022
VidMob *New York, NY*

- Built a deep learning model to score creative ad performance, improving scoring accuracy by 25%.
- Integrated predictive modeling and recommendation engines into an AI-driven ad-tech platform.
- Conducted model evaluation pipelines to assess ad engagement prediction, resulting in better client satisfaction by 5%.

Student Researcher Jul 2020 – January 2023
University of California, Los Angeles(UCLA) *Los Angeles, CA*

- Co-authored paper (SIGGRAPH 2022) on multimodal sensing using radar and RGB data for equitable rPPG.
- Proposed a Shift-Robust Loss Function for pulse prediction, decreasing error by 40%.
- Built ML pipelines to mitigate skin-tone bias using sensor fusion, improving fairness and accuracy by 75%.

Software Engineer Jun 2019 – Jul 2021
Accenture *Bengaluru, India*

- Improved pharma sales data analysis via automated data processing pipelines, reducing manual time by 50%.
- Boosted competitor match rate by 20% via advanced analytics, earning cross-region recognition.
- Participated in Accenture's Global Innovation Challenge as a semi-finalist.

Education

University of California, Los Angeles(UCLA) Sept 2021 - Oct 2023
MS in Electrical and Computer Engineering [GPA: 3.97/4.0] *Los Angeles, CA*

PES University Aug 2015 - Aug 2019
BTech in Electronics and Communication Engineering *Bangalore, India*

Projects

- **Automatic Garbage Segregator:**
  1) Engineered a crane that could segregate waste into biodegradable, non-biodegradable, and electronic waste with an accuracy of 95.18%.
  2) Tested various Feature Extraction techniques like PCA, LDR, and Convolutional Neural Networks.
- **Multi-Class EEG Motor Imagery Classification Using Deep Learning Architectures**
  1) Implemented various deep learning techniques, including CNNs, LSTMs, RNNs, VAEs, Transformers and attention to achieve multi-class classification accuracy of EEG signals for motor imagery tasks.
  2) Demonstrated the potential of deep learning techniques for EEG signal analysis by achieving a classification accuracy rate of 75% on the entire dataset.
- **Detecting Pulse from Head Movement:** I replicated the paper "Detecting Pulse from Head Movement" by Guha Balakrishnan, Fredo Durand, John Guttag.The code was able to detect the Heartbeat with an error of around 2-5%. The code was done in Python and it took around 2.5 weeks to complete

Publication & Certifications

SIGGRAPH 2022 Blending camera and 77 GHz radar sensing for equitable, robust plethysmography

Extracurricular Activities

- **Eyantra 2016 Robotics Competition**: Coached a team of 3 students to the semi-finals in a National Level Robotics Competition. Managed and delegated tasks effectively to meet deadlines. Designed a robot with navigation capability and an arm for object pickup.
- Professional Bharatanatyam dancer (Top 1% in Senior Exam)
- Presented at IEEE Symposium Series on Computational Intelligence
- Member, Center for Intelligent Systems at PES University
- Operations Lead, Epsilon 2016 (2000+ participants, 100+ events)